

Data Structure(CSL202)

By-
Dr. Sk Subidh Ali (subidh@iitbhilai.ac.in)



Infix to Postfix

Reverse Polish Notation or Postfix Expressio

- The operator comes after its operands.

Example-

Infix notation	(A + B) - (C * D)
Postfix	AB+ CD* -

Operator	Precedence Rules	Associativity
^	Highest	Right to Left
*, /	Higher	Left to Right
+, -	Lower	Left to Right

Infix to Postfix

1. Push '(' onto STACK.

2. Append ')' to the end of Q.

3. Repeat for each symbol in Q (from left to right):

   a. If symbol is an operand:

      → Append to P.

   b. If symbol is '(':

      → Push onto STACK.

   c. If symbol is an operator ⊗:

      → While top of STACK is an operator with precedence ≥ ⊗:

         Pop from STACK and append to P.

      → Push ⊗ onto STACK.

   d. If symbol is ')':

      → While top of STACK ≠ '(':

         Pop from STACK and append to P.

      → Pop '(' from STACK (do not append to P).

4. End loop when STACK is empty.

5. Exit.

Q is the arithmetic expression
P is the Postfix version of Q

Example

- **Infix expression:**

  d/(e+f)+b*c

- **Postfix Expression:**

  def+/ b c * +

Symbol Scanned	STACK	Expression P
d	(	d
/	(/	d
(	(/ (	d
e	(/ (	de
+	(/ (+	de
f	(/ (+	def
)	(/	def+
+	(+	Def+ /
b	(+	Def+ / b
*	(+ *	def+ / b
c	(+ *	def+/ b c
)		def+/ b c * +

Infix to Postfix Complexity

- **Time Complexity:**
  - O(n) in all cases.
  - Where n is the size of the infix expression
- **Space Complexity:**

  O(n) for storing:

  - Output postfix expression (P).
  - Stack (for operators and parentheses).



Postfix Expression Evaluation

Postfix Expression Evaluation

1. Append ')' to end of P as a sentinel.

2. Scan P from left to right until ')' is encountered:

   a. If element is an operand:

      → Push onto STACK.

   b. If element is an operator (⊗):

      → Pop A ← top of STACK

      → Pop B ← next-to-top of STACK

      → Compute RESULT ← B ⊗ A

      → Push RESULT onto STACK

3. After loop, VALUE ← Top of STACK

4. Exit.

Example

❖ **Postfix Expression:**

8 2 3 ^ / 2 3 * + 5 1 * -

= 2

Symbol Scanned	STACK
8	**8**
2	**8 2**
3	**8 2 3**
^	**8 8**
/	**1**
2	**1 2**
3	**1 2 3**
*	**1 6**
+	**7**
5	**7 5**
1	**7 5 1**
*	**7 5**
-	**2**

Postfix Expression Evaluation

- **Time Complexity:**
  - O(n) in all cases.
  - Where n is the size of the infix expression

Space Complexity:

- **O(n) for the stack:**
  - All operands may be on stack simultaneously in certain cases.